Chain-of-Cooking and CookViz: Bidirectional Thought Guidance for Cooking Process Visualization

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Abstract

Cooking process visualization aims to generate an image for each step of a recipe while preserving ingredient identity, reflecting state changes introduced by actions such as cutting, mixing, or frying, and maintaining coherence over the entire procedure. Existing text-to-image models are optimized for single images, while prior procedural generation methods either target open-domain stories or rely on limited cooking data that does not provide sufficiently clean, step-aligned supervision. We present Chain-of-Cooking (CoCook), a diffusion-based framework that augments a frozen Stable Diffusion v1.5 backbone with three complementary mechanisms: dynamic patch selection from historical frames, semantic evolution of step prompts in latent space, and bidirectional thought guidance during denoising. To support evaluation, we introduce CookViz, a public English benchmark with 40,362 high-resolution, step-aligned cooking images paired with recipe instructions and organized with a 9:1 train/test split. Across both RecipeQA and CookViz, CoCook outperforms fine-tuned Stable Diffusion, SDXL, StoryGen, TIP, and ImageRAG on FID, CLIP-based alignment, DreamSim, and human judgments of semantic consistency and frame coherence. On CookViz, CoCook achieves 12.28 FID, 0.218 CLIP-T, 0.820 CLIP-I, and 0.5851 DreamSim, together with the best human ratings of 4.22 for consistency and 4.30 for coherence. Ablations show that patch-level retrieval, semantic evolution, and the semantic consistency loss each contribute to the final performance.

CCS Concepts

• **Computing methodologies** → **Neural networks**; **Image generation**; *Computer vision tasks.*

Keywords

cooking process visualization, diffusion models, procedural image generation, retrieval-augmented generation, benchmark datasets

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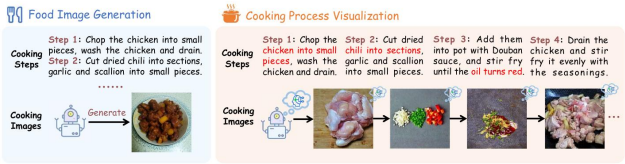

Figure 1: Comparison between traditional food image generation and cooking process visualization. The latter requires coherent multi-step rendering with evolving ingredient appearance and correct procedural order.

1 Introduction

Text-to-image generation has progressed rapidly with large generative models such as DALL-E 3, SD3, and SDXL, which can synthesize high-quality images from natural language descriptions [1, 2, 13]. Yet cooking process visualization remains substantially harder than standard text-to-image generation because the target is not a single image but an ordered sequence of intermediate states. A successful model must preserve the identity and appearance of ingredients across steps, update those appearances when the procedure changes them, and remain faithful to the current instruction without drifting toward earlier or later stages.

This setting differs from traditional food generation, which usually focuses on a final plated dish or a recipe-level image [5, 11, 25]. It is also distinct from open-domain story visualization and multimodal planning, where the dominant challenges are character consistency or cross-modal planning rather than fine-grained physical state transitions of food [9, 10, 15]. Figure 1 contrasts the standard food-image setting with cooking process visualization.

Progress on this problem is additionally constrained by data. RecipeQA is public and multimodal, but it was created for comprehension rather than generation [20]. Other cooking resources are either non-public, in another language, or designed for workflow reasoning instead of step-by-step image synthesis [6, 12, 23]. As a result, it is difficult to both train models that respect intermediate visual transformations and compare methods under a common protocol.

We address both limitations with a method contribution and a benchmark contribution. Our method, Chain-of-Cooking (CoCook), builds on Stable Diffusion v1.5 and injects step-aware visual evidence from prior frames into the denoising process. The first component, Dynamic Patch Selection (DPS), retrieves patch tokens from historical cooking frames using the current instruction as a query. The second component, Semantic Evolution (SE), initializes learnable latent prompts for the current step and fuses them with

textual conditioning. The third component, Bidirectional Chain-of-Thought (CoT) Guidance, modulates latent prompts differently in the forward and reverse diffusion directions and regularizes them with semantic consistency objectives. The overall design is shown in Figure 2.

Our benchmark contribution is CookViz, a public dataset of 40,362 high-resolution English cooking frames with aligned step descriptions. Each sample contains at least three steps, enabling evaluation of temporal coherence and appearance preservation across meaningful transformations. Beyond reporting standard automatic metrics, we evaluate semantic consistency and frame coherence through a 50-participant human study.

The contributions of this paper are threefold:

- We introduce CoCook, a diffusion framework for cooking process visualization that combines patch-level retrieval from prior frames, latent prompt evolution, and bidirectional thought guidance.
- We introduce CookViz, a public benchmark for step-aligned cooking visualization, together with an evaluation protocol that combines FID, CLIP-based alignment, DreamSim, and human judgments.
- We provide comprehensive experiments on RecipeQA and CookViz showing that CoCook consistently improves visual fidelity, text alignment, and sequence coherence over strong procedural and retrieval-based baselines.

2 Related Work

2.1 Food and Procedural Image Generation

Early food-image generation work concentrated on recipe-level or dish-level synthesis rather than intermediate procedural states. CookGAN models causal interactions between recipe text and food appearance, while FoodFusion and ClusDiff adapt modern generative modeling to the food domain [5, 11, 25]. R2GAN explores cross-modal recipe retrieval with image generation as an auxiliary capability [26]. More recently, CookingDiffusion explicitly targets procedural cooking image generation, highlighting the importance of modeling state transitions across steps [18]. Our work shares this procedural focus, but differs in two ways: we use retrieval over historical visual evidence at the patch level, and we introduce a public benchmark specifically designed for step-aligned cooking visualization.

2.2 Sequential and Story Visualization

Sequence generation has also been studied in story visualization and multimodal planning. Make-a-Story and StoryGen use diffusion-based memory or context modules to improve consistency across generated frames [9, 15]. TIP addresses multimodal procedural planning by jointly generating text and images for plans [10]. These methods demonstrate the value of conditioning on prior context, but they are not specialized for the appearance-preserving, transformation-heavy nature of cooking, where small local changes such as chopping, mixing, or browning strongly affect correctness.

2.3 Retrieval and Reasoning for Image Generation

Retrieval-augmented generation has become an effective way to inject external evidence into generative models. ImageRAG dynamically retrieves reference images to improve rare or fine-grained synthesis [17]. In parallel, reasoning-oriented work has explored chain-of-thought style decomposition for text-to-image systems and multimodal generation [4, 19, 21, 22, 24]. Our use of "thought guidance" is not language-chain generation; instead, it denotes a bidirectional latent-conditioning strategy that encourages current-step semantics to remain stable across denoising while still adapting to the procedural context.

2.4 Cooking and Recipe Benchmarks

RecipeQA provides public multimodal recipes with aligned text and images, but it was introduced for comprehension tasks [20]. Recipe1M is large-scale and valuable for recipe-image retrieval, but it is not a benchmark for step-by-step generation [16]. Cookpad, MIAIS, and MM-ReS contain useful cooking signals, yet access, language coverage, or task design limit their suitability for evaluating public cooking process visualization systems [6, 12, 23]. CookViz is designed to fill this gap by prioritizing public availability, English instructions, and step-aligned frames for generation and evaluation.

3 CookViz Benchmark

3.1 Design Goals

CookViz is designed to evaluate whether a model can render an entire cooking procedure rather than a single final dish. This requires three properties from the benchmark: aligned text-image pairs for each step, sufficient sequence length to expose temporal drift, and image quality high enough for both training and human evaluation. We therefore retain only English recipes with at least three aligned steps and high-resolution images.

3.2 Collection and Cleaning

CookViz is constructed from public cooking websites that provide ordered step descriptions together with corresponding images. The collection pipeline preserves the original step order of each recipe and keeps only complete step-image pairs. Text cleaning removes emojis and irrelevant characters with regular expressions. To improve visual quality, OCR is used to identify watermarks and Stable Diffusion inpainting is used to remove detected marks. Recipes with incomplete processes or low-quality images are discarded.

3.3 Task Definition and Quality Control

Each benchmark example is an ordered sequence of step descriptions and images. Models are evaluated on their ability to generate the current step while remaining visually compatible with prior states. Requiring at least three steps filters out near-static examples and makes temporal coherence measurable. Preserving the original recipe order avoids heuristic realignment noise. Watermark removal is also important in this setting because repeated artifacts can be copied across steps and create spurious consistency without improving procedural correctness.

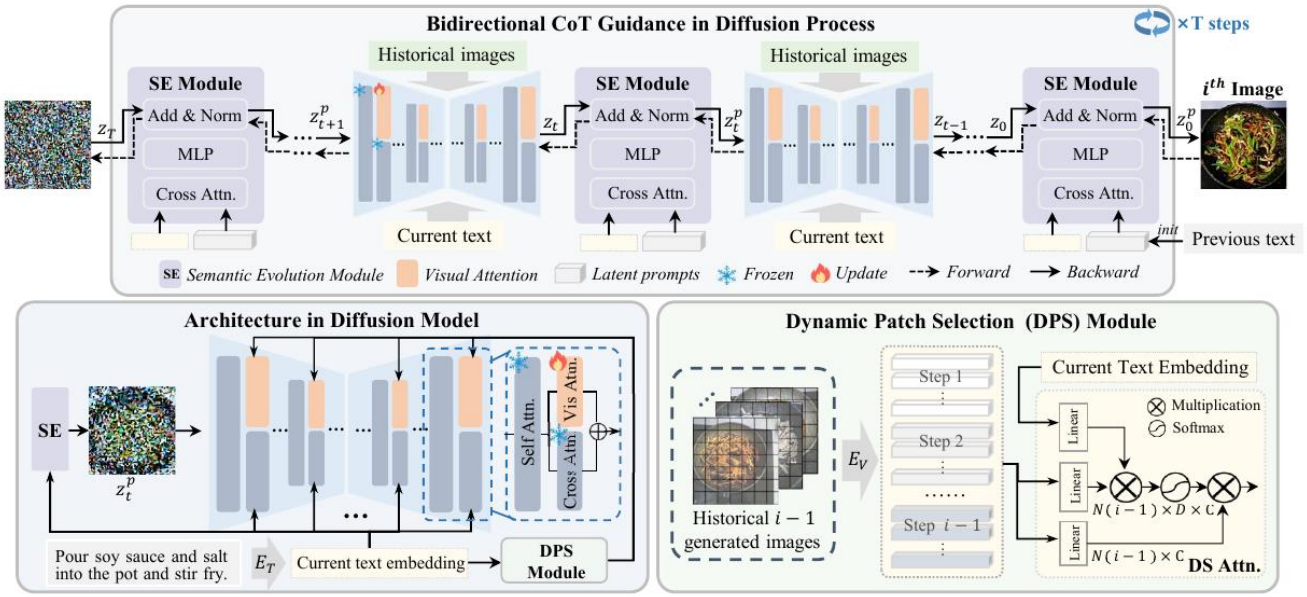

Figure 2: Overview of CoCook. The model augments a Stable Diffusion v1.5 backbone with Dynamic Patch Selection from historical frames, Semantic Evolution of latent prompts, and Bidirectional CoT Guidance during denoising.

Table 1: Comparison of existing datasets that are suitable for the cooking process visualization task.

Dataset	Public	#Frames	Language
Cookpad [6]	False	1,642,450	Japanese
MIAIS [23]	False	121,547	Japanese
MM-Res [12]	False	227,082	English
RecipeQA[20]	True	approx. 20,000	English
CookViz	True	40,362	English

3.4 Statistics and Comparison

The resulting benchmark contains 40,362 frames and follows a 9:1 train/test split. All samples are in English and publicly available. Table 1 compares CookViz with prior datasets relevant to cooking visualization. Compared with RecipeQA, CookViz is larger in terms of step-aligned frames for this task. Compared with Cookpad, MIAIS, and MM-ReS, it emphasizes public accessibility and direct suitability for procedural image generation.

3.5 Evaluation Protocol

We evaluate generated sequences with four automatic metrics and one human study. FID measures distributional fidelity [8]. CLIP-T measures alignment between the generated image and the current step text using CLIP features [7, 14]. CLIP-I measures similarity between a generated step and its reference image using the same representation [14]. DreamSim assesses perceptual similarity beyond low-level pixel statistics [3]. Following the benchmark protocol, we also conduct a 50-participant human study that scores semantic consistency and frame coherence on a 1–5 scale. Consistency measures whether a frame matches the current instruction; coherence measures whether consecutive frames form a plausible visual process.

The evaluation suite is deliberately complementary. FID measures global image quality, CLIP-T measures step-level semantic grounding, CLIP-I measures faithfulness to the paired reference frame, and DreamSim adds a perceptual criterion that is less tied to low-level pixel differences. Together with human ratings, these metrics distinguish realistic but procedurally incorrect images from genuinely coherent cooking sequences. We report sequence scores after generating complete recipes in order, so every metric reflects both per-step quality and the accumulation of temporal drift over the full procedure.

4 Method

4.1 Problem Formulation

Given a recipe with ordered instructions $\{s_1, \ldots, s_T\}$, the goal is to generate a corresponding image sequence $\{\hat{x}_1, \ldots, \hat{x}_T\}$ such that each $\hat{x}_t$ is faithful to step s_t and coherent with the preceding frames. CoCook starts from a Stable Diffusion v1.5 backbone and freezes the base text encoder, VAE, image encoder, and most UNet weights. Training updates only the additional cross-attention pathways used to integrate historical visual evidence and evolved latent prompts.

4.2 Dynamic Patch Selection

The first challenge is selecting useful visual evidence from earlier steps. Whole-frame retrieval is often too coarse because the current step may only depend on a subset of ingredients or local visual states. CoCook therefore encodes all prior frames with a CLIP vision

Table 2: Diagnostic dimensions covered by CookViz.

Aspect	Typical failure	Primary signals
Identity	ingredient color or shape changes without cause	CLIP-I, coherence
Action	generated frame ignores verbs such as chop, mix, or fry	CLIP-T, consistency
Transition	adjacent steps are nearly static or change too abruptly	consistency, coherence
Realism	artifacts dominate otherwise aligned steps	FID, DreamSim

encoder and removes the global classification token, keeping patch-level tokens only. For step t, let $H_t \in \mathbb{R}^{L\times d}$ be the text embedding of the current instruction and let $V_{<t} \in \mathbb{R}^{M\times d}$ be the concatenation of patch tokens from historical frames. The retrieval weights are computed by text-to-patch attention:

$$R_t = \text{softmax}\left(\sum_{i=1}^{L} \frac{H_t^{(i)} V_{<t}^{\top}}{\sqrt{d}}\right) V_{<t}, \tag{1}$$

where R_t is the retrieved visual context injected into the UNet through learnable cross-attention layers. This design matches the released implementation, which forms historical CLIP patch tokens and uses the current text embedding to produce attention-weighted reference features.

4.3 Semantic Evolution

Text alone is insufficient to preserve appearance continuity over a long sequence. CoCook therefore augments each step with a learnable latent prompt. Given the current text embedding and the current latent z_t, a latent-prompt block cross-attends the prompt to the text embedding and projects the result to the latent tensor. This yields a step-specific conditioning signal that is added directly to the denoising latent. Intuitively, Semantic Evolution turns a static textual step description into a step-aware latent state that can be aligned with both image evidence and diffusion noise.

This module is lightweight by design. In the released implementation, the VAE, text encoder, CLIP image encoder, and most UNet weights remain frozen, while training is concentrated on the newly introduced cross-attention pathways. This parameter-efficient design is well matched to recipe datasets, which are far smaller than general web-scale text-to-image corpora.

4.4 Bidirectional CoT Guidance

The diffusion training loop uses two latent prompts with opposite timestep modulation. One prompt is emphasized more strongly at earlier reverse denoising stages, while the other is emphasized more strongly when constructing the forward noising path. This bidirectional design encourages the model to preserve stable semantics throughout denoising rather than depending on a single conditioning pathway. In the released implementation, the text encoder states are concatenated with the time-modulated latent prompt before being passed to the UNet, and the historical visual context R_t is supplied as an additional cross-frame feature.

Generation proceeds autoregressively over recipe steps. The first step uses a null visual reference. Later steps encode previously generated frames, retrieve the most relevant historical patch evidence with DPS, and condition the current denoising process on that evidence. This setup makes the task nontrivial: the model must preserve appearance when appropriate, but still change it when the current instruction requires a new cooking state.

4.5 Training Objective

CoCook optimizes the standard denoising loss together with semantic consistency regularization for both prompt directions. Let ϵ denote the sampled noise, $\hat{\epsilon}$ the UNet prediction, and P_t^b, P_t^f the two latent prompts. The training objective is

$$\mathcal{L} = \|\hat{\epsilon} - \epsilon\|_2^2 + \lambda\left(1 - \cos(P_t^b, H_t)\right) + \lambda\left(1 - \cos(P_t^f, H_t)\right), \tag{2}$$

where $\lambda = 0.1$ in the training code. We use 512×512 images, a learning rate of 10^{-5}, and 50 diffusion inference steps in the released setup.

5 Experiments

5.1 Experimental Setup

We evaluate CoCook on RecipeQA and CookViz. RecipeQA serves as a public multimodal recipe benchmark with approximately 20K recipes and aligned instructional images [20]. CookViz is the benchmark introduced in this paper. All methods generate complete image sequences from ordered recipe steps.

We compare against five baselines: fine-tuned Stable Diffusion v1.5, fine-tuned SDXL [13], StoryGen [9], TIP [10], and ImageRAG [17]. The first two are strong diffusion baselines, StoryGen is a sequence generation baseline, TIP is a multimodal planning method, and ImageRAG represents retrieval-augmented image generation. We report FID, CLIP-T, CLIP-I, DreamSim, and human ratings of consistency and coherence.

All methods are evaluated in the same ordered generation setting. The diffusion-based systems operate at 512×512 resolution, and the released CoCook code uses 50 denoising steps with guidance scale 7.5. We retain the original processed train/test splits and report the benchmark numbers provided with the release artifacts.

5.2 Main Results

Table 4 reports the quantitative and human evaluation results. CoCook is the best model on every automatic metric reported for both datasets. On RecipeQA, CoCook improves FID from the strongest baseline score of 16.81 to 12.43 and obtains the best CLIP-T, CLIP-I, and DreamSim values. On CookViz, CoCook achieves 12.28 FID, 0.218 CLIP-T, 0.820 CLIP-I, and 0.5851 DreamSim, outperforming the best competing baseline by a clear margin on FID and by smaller but consistent margins on semantic alignment.

Table 3: Margin of CoCook over the strongest baseline on representative metrics.

Dataset	FID gain	CLIP-T gain	Coher. gain
RecipeQA	4.38	0.001	0.16
CookViz	1.52	0.003	0.28

Human evaluation follows the same trend. CoCook reaches 3.84 consistency and 3.86 coherence on RecipeQA, and 4.22 consistency and 4.30 coherence on CookViz. These scores exceed all baseline models, indicating that the gains are perceptible to annotators and not limited to automatic metrics.

Two trends stand out. First, method ranking is broadly consistent across automatic and human evaluation, which strengthens the claim that the improvements are not metric-specific. Second, the advantages of CoCook remain clear on CookViz, suggesting that the cleaner step alignment and larger frame count of the new benchmark make sequence-level modeling benefits more visible rather than artificially inflating all methods.

Table 3 shows that CookViz is especially diagnostic for sequence quality: its largest relative improvement appears on human coherence rather than FID. This pattern complements the absolute scores in Table 4 and suggests that the benchmark is sensitive to cross-step consistency gains instead of rewarding only single-frame realism.

5.3 Qualitative Comparison

Figure 3 shows representative generations. Fine-tuned single-image baselines often produce visually plausible frames but drift in ingredient appearance across steps. StoryGen improves sequence-level consistency but is less specialized for cooking transformations. TIP better follows the high-level procedure but can miss visual details. ImageRAG helps preserve visual evidence but remains less step-specific than CoCook. In contrast, CoCook produces sequences that better preserve ingredient identity while still reflecting action-induced changes such as cutting, mixing, or heating.

6 Analysis

6.1 Component Ablations

Table 5 ablates the main components of CoCook. Removing DPS causes the largest degradation in FID on both datasets, from 12.43 to 14.05 on RecipeQA and from 12.28 to 13.76 on CookViz, while also reducing CLIP-I. This confirms that retrieving localized historical evidence is central to preserving appearance continuity. Removing Semantic Evolution degrades CLIP-T more noticeably, suggesting that latent prompt evolution primarily improves step-level semantic alignment. Removing the semantic consistency loss also hurts both CLIP-T and CLIP-I, showing that explicit regularization is useful even when the denoising objective is already present.

The same table also compares whole-frame and patch-level retrieval. Patch retrieval achieves better FID and CLIP-T than frame retrieval on both datasets, while matching or improving CLIP-I. This supports the design choice behind DPS: in cooking, only a subset of prior visual evidence is usually relevant to the current step.

6.2 Visual Analysis of Components

Figure 4 visualizes the effect of removing DPS, SE, and the semantic consistency loss. The failures are consistent with the quantitative ablations: without DPS, the model loses local ingredient fidelity; without SE, it more often drifts away from the current instruction; without semantic consistency loss, sequences remain plausible but less stable.

Figure 5 further compares intermediate decoded images during denoising. Relative to the Stable Diffusion baseline, CoCook reaches recognizable, step-relevant structure earlier in the denoising trajectory, which is consistent with the intended role of the bidirectional thought guidance.

6.3 Benchmark Value

CookViz is not only a training resource but also a diagnostic benchmark. Compared with RecipeQA, it yields stronger human scores for the best model and offers more high-resolution frames, making it easier to separate methods that merely produce plausible food images from methods that truly maintain procedural coherence. The fact that all baselines still lag clearly behind ground truth on human ratings indicates that the benchmark remains challenging and leaves room for future progress.

CookViz is also discriminative. On the new benchmark, differences between baselines remain visible across FID, CLIP-T, CLIP-I, DreamSim, and human scores instead of collapsing into a narrow band. StoryGen is relatively competitive on coherence, TIP is relatively competitive on text alignment, and ImageRAG benefits from retrieval but still trails CoCook. This spread is useful because it reveals which design choices primarily improve realism, which improve semantic grounding, and which improve sequence stability. Table 2 summarizes this intended diagnostic role. In practice, the benchmark exposes different error modes rather than rewarding one narrow notion of quality: retrieval-heavy methods help identity preservation, planning-oriented methods help action grounding, and CoCook is strongest because it improves these dimensions jointly.

7 Conclusion

We presented CoCook, a diffusion-based framework for cooking process visualization that combines dynamic patch-level retrieval, latent semantic evolution, and bidirectional thought guidance. We also introduced CookViz, a public benchmark with 40,362 English step-aligned cooking frames and a unified evaluation protocol combining automatic metrics and human judgments. Across RecipeQA and CookViz, CoCook consistently outperforms strong fine-tuned, sequential, planning, and retrieval-based baselines. The results suggest that explicitly combining historical visual evidence with step-aware latent prompt guidance is an effective direction for procedural image generation, and that CookViz provides a meaningful benchmark for measuring future progress on this task.

Table 4: Quantitative evaluations and human evaluations (Consis., Coher.) on the RecipeQA and CookViz datasets.

Model	RecipeQA						CookViz					
	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑
GT	-	0.207	1	1	4.12	3.98	-	0.215	1	1	4.76	4.80
fine-tuned SD	18.35	0.200	0.733	0.6145	3.70	3.44	13.80	0.210	0.813	0.6150	3.90	3.84
fine-tuned SDXL [13]	18.76	0.202	0.730	0.6308	3.78	3.46	13.81	0.212	0.813	0.6163	3.96	3.84
StoryGen [9]	17.36	0.204	0.737	0.6105	3.62	3.62	16.77	0.213	0.818	0.6065	3.84	4.02
TIP [10]	18.69	0.208	0.734	0.62440	3.60	3.38	14.65	0.215	0.814	0.6127	3.80	3.96
ImageRAG [17]	16.81	0.204	0.735	0.6120	3.72	3.70	13.91	0.212	0.815	0.6100	4.04	3.98
CoCook	**12.43**	**0.209**	**0.739**	**0.5941**	**3.84**	**3.86**	**12.28**	**0.218**	**0.820**	**0.5851**	**4.22**	**4.30**

Figure 3: Qualitative comparison on procedural cooking image generation. CoCook better preserves ingredient appearance across steps while matching the current instruction more faithfully than fine-tuned SD, SDXL [13], StoryGen [9], and TIP [10].

Table 5: Ablation experiments of key components and image selection methods on the RecipeQA and CookViz datasets.

Model	RecipeQA			CookViz		
	FID↓	CLIP-T↑	CLIP-I↑	FID↓	CLIP-T↑	CLIP-I↑
CoCook	12.43	0.209	0.739	12.28	0.218	0.820
w/ o DPS	14.05	0.207	0.732	13.76	0.217	0.817
w/ o SE	12.92	0.204	0.736	12.89	0.215	0.819
w/ o $\mathcal{L}_{CS}$	12.60	0.205	0.735	12.41	0.215	0.817
w/ frame	12.84	0.204	0.732	12.72	0.216	0.820
w/ patch	12.43	0.209	0.739	12.28	0.218	0.820

Figure 4: Visual ablation of CoCook. Removing DPS, SE, or the semantic consistency loss degrades either appearance preservation or step-level faithfulness.

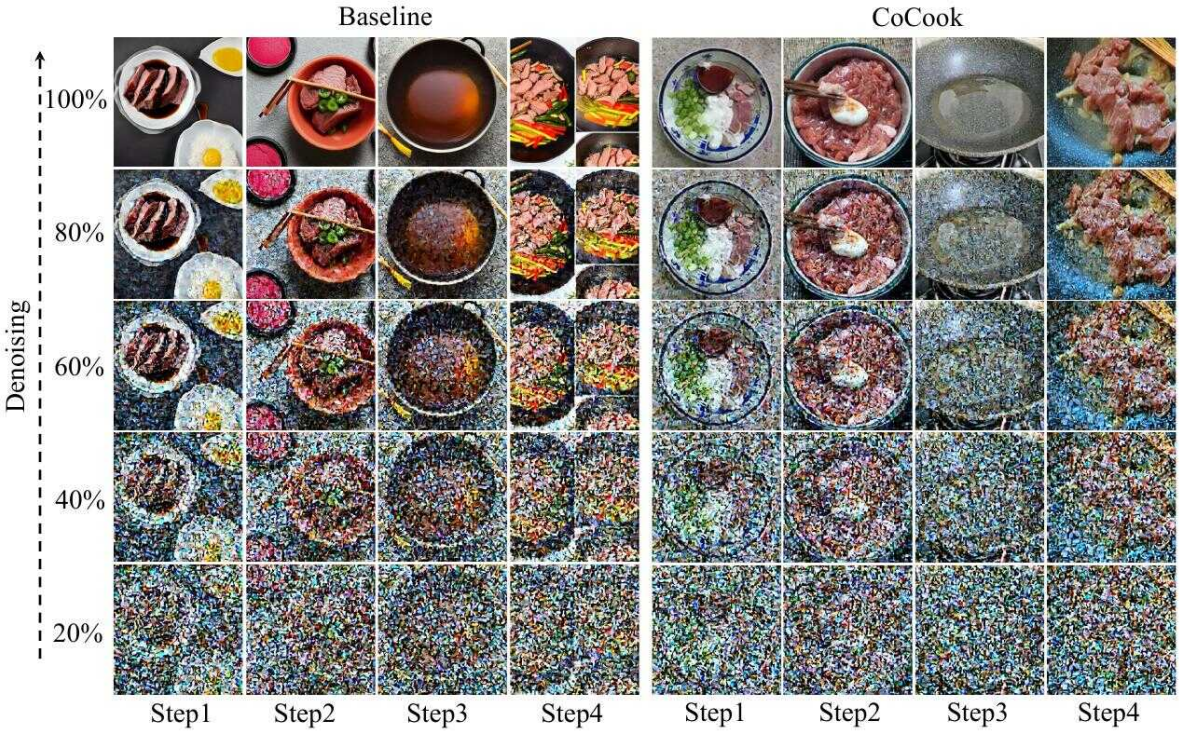

Figure 5: Intermediate decoded images during denoising for the Stable Diffusion baseline and CoCook. CoCook shows more stable step-aware structure through the denoising process.

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